

Name: Arijun Patel Date: 09.27.24

Point values are assigned for each question.

Points earned: ____ / 100, = ____ %

1. Find a tight upper bound for $f(n) = n^5 + 8$. Write your answer here: $O(n^5)$ (4 points)

Prove your answer by giving values for the constants c and n_0 . Choose the smallest integer value possible for c . (4 points)

$$n_0 = 2, \quad C = 2$$

See separate page for solution

2. Find an asymptotically tight bound for $f(n) = n^3 + 5n$. Write your answer here: $\Theta(n^3)$ (4 points)

Prove your answer by giving values for the constants c_1 , c_2 , and n_0 . Choose the tightest integer values possible for c_1 and c_2 . (6 points)

$$n_0 = 3, \quad c_1 = 1, \quad c_2 = 2$$

See separate page for solution

3. Is $2n^2 - n \in \Omega(n^3)$? Circle your answer: yes (no) (2 points)

If yes, prove your answer by giving values for the constants c and n_0 . Choose the smallest integer value possible for c . If no, derive a contradiction. (4 points)

See Proof

See separate page for solution r

4. Write the following asymptotic efficiency classes in **increasing** order of magnitude.

~~$O(\lg n^2)$~~ , ~~$O(0.01^n)$~~ , ~~$O(20!n)$~~ , ~~$O(n \lg n)$~~ , ~~$O(n)$~~ , ~~$O(n!)$~~ , ~~$O(n^8)$~~ , ~~$O(\lg \lg n)$~~ , ~~$O(n^n)$~~ , ~~$O(n^{20} \lg n)$~~ (2 points each)

$O(0.01^n)$, $O(\lg \lg n)$, $O(\lg n^2)$, $O(n)$, $O(20!n)$, $O(n \lg n)$, $O(n^8)$, $O(n^2 \lg n)$, $O(n!)$, $O(n^n)$

Additional work on separate page

1) Tight upper bound for $f(n) = n^5 + 8$

$O(g(n)) = \{f(n) : \text{there exists positive constants } c \text{ and } n_0 \text{ such that } 0 \leq f(n) \leq cg(n) \forall n \geq n_0\}$

$$0 \leq f(n) \leq cg(n) \quad \forall n \geq n_0$$

$$0 \leq n^5 + 8 \leq cg(n)$$

by looking at the function, the dominant term is n^5 , so we can use $O(n^5)$ as a tight upper bound

$$n^5 + 8 \leq cn^5 \quad \forall n \geq n_0$$

$$n^5 + 8 \leq n^5 + n^5 \quad n \geq 8$$

$$n^5 + 8 \leq 2n^5 \quad n \geq 8$$

$$2n^5 \leq cn^5$$

$$c = 2$$

$$\hookrightarrow O(n^5)$$

$$n=1 \Rightarrow 9 \leq 2 \quad \times$$

$$n=2 \Rightarrow 40 \leq 64 \quad \checkmark$$

but, we can find a smaller $n_0 \Rightarrow$

$$c = 2, n_0 = 2$$

2) Asymptotically tight bound for $f(n) = n^3 + 5n$

$\Theta(g(n)) = \{f(n) : \text{there exists positive constants } c_1, c_2, \text{ and } n_0 \text{ such that } 0 \leq c_1 g(n) \leq f(n) \leq c_2 g(n) \forall n \geq n_0\}$

$$0 \leq c_1 g(n) \leq f(n) \leq c_2 g(n) \quad \forall n \geq n_0$$

$$c_1 g(n) \leq n^3 + 5n \leq c_2 g(n)$$

$\underbrace{\hspace{1cm}}$

$\underbrace{\hspace{1cm}}$

by looking at $f(n)$, the dominant term is n^3 , so a tight bound for $f(n)$ is $\Theta(n^3)$

$$\hookrightarrow \Theta(n^3)$$

$$c_1 n^3 \leq n^3 + 5n \leq c_2 n^3 \quad \forall n \geq n_0$$

$$n^3 + 5n \leq c_2 n^3 \quad \forall n \geq n_0 \quad \leftarrow c_1 n^3 \leq n^3 + 5n$$

$$n^3 + 5n \leq n^3 + n^3$$

$$n^3 + 5n \leq 2n^3$$

$$2n^3 \leq c_2 n^3$$

$$c_2 = 2$$

now to find $n_0 \Rightarrow$

$$n=1 \Rightarrow 6 \leq 2 \quad \times$$

$$n=2 \Rightarrow 18 \leq 16 \quad \times$$

$$n=3 \Rightarrow 42 \leq 54 \quad \checkmark$$

$$c_2 = 2, n_0 = 3$$

$$c_1 n^3 \leq n^3 + 5n \quad \forall n \geq n_0$$

Smallest n_0 is 1 to satisfy

$$c_1 \leq \frac{n^3}{n^3} + \frac{5n}{n^3}$$

$$c_1 \leq 1 + \frac{5}{n^2} \quad \text{as } n \rightarrow \infty \quad \frac{5}{n^2} = 0$$

$$c_1 \leq 1 + 0$$

$$c_1 = 1$$

$$c_1 = 1, c_2 = 2, n_0 = 3$$

3) is $2n^2 - n \in \Omega(n^3)$?

$$2n^2 - n \notin \Omega(n^3)$$

$$c_1 n^3 \leq 2n^2 - n$$

$\hookrightarrow n^3$ will grow at a faster rate than $2n^2 - n$ as $n \rightarrow \infty$, so it is not possible for $n^3 \leq 2n^2 - n$

$$c_1 n^3 \leq 2n^2 - n$$

$$c_1 \leq \frac{2n^2 - n}{n^3}$$

$$c_1 \leq \frac{2}{n} - \frac{1}{n^2}$$

as $n \rightarrow \infty$ both terms

become zero

which cannot be greater than or equal to a positive constant c

$$c_1 n^3 \leq 2n^2 - n$$

$$c_1 n^3 \leq 2n^3 - n^2$$

$$c_1 n^3 \leq n^2$$

$$n \leq \frac{1}{c}$$

as $n \rightarrow \infty$, there exists no positive value for

c

where

n is

bounded by $\frac{1}{c}$.

4)

$$\lim_{n \rightarrow \infty} \frac{n^8}{n \lg n} = \frac{n^7}{\lg n} = \infty \quad O(n^8) \text{ has larger order than } O(n \lg n)$$

- First, we can see that $O(0.01^n)$ has low magnitude compared to the other complexities
- we can also see that $O(n^n)$ has high magnitude compared to the other complexities

$$\lim_{n \rightarrow \infty} \frac{t(n)}{g(n)} = \begin{cases} 0 & \text{implies that } t(n) \text{ has a smaller order of growth than } g(n), \\ c & \text{implies that } t(n) \text{ has the same order of growth as } g(n), \\ \infty & \text{implies that } t(n) \text{ has a larger order of growth than } g(n).^3 \end{cases}$$

lets compare $\underbrace{O(\lg n^2)}_{t(n)}$ and $\underbrace{O(n \lg n)}_{g(n)}$

$$\lim_{n \rightarrow \infty} \frac{\lg n^2}{n \lg n} = \frac{2 \lg n}{n \lg n} = \frac{2}{n} = 0 \quad O(\lg n^2) \text{ smaller than } O(n \lg n)$$

$$\lim_{n \rightarrow \infty} \frac{O(\lg n^2)}{\lg \lg n} = \frac{2 \lg n}{\lg \lg n} = \infty \quad O(\lg n^2) \text{ larger growth than } O(\lg \lg n)$$

$$\lim_{n \rightarrow \infty} \frac{O(n \lg n)}{\lg \lg n} = \infty \quad O(n \lg n) \text{ larger growth than } O(\lg \lg n)$$

$O(n!)$ vs $O(n^{20} \lg n)$

$$\lim_{n \rightarrow \infty} \frac{n!}{n^{20} \lg n} = \frac{\sqrt{2\pi n} \left(\frac{n}{e}\right)^n}{n^{20} \lg n} \xrightarrow{\text{stirling's formula}} \lim_{n \rightarrow \infty} \frac{\sqrt{2\pi n} n^n}{n^{20} \lg n e^n} = \infty$$

$O(n!)$ larger magnitude than $O(n^{20} \lg n)$

5. Give the complexity of the following methods. Choose the most appropriate notation from among O , Θ , and Ω . (8 points each)

```
int function1(int n) {
    int count = 0; runs in constant time  $C_0$ 
    for (int i = 0; i <= n/2; i++) { runs  $n/2$  times
        for (int j = 1; j <= n; j *= 2) { runs  $\log_2 n$  times  $O(\log_2 n)$ 
            count++; runs in constant time  $C_1$ 
        }
    }
    return count;
}
```

complexity = $C_0 + \frac{n}{2} \times \log_2 n \times C_1$
 $\frac{n}{2} \log_2 n \Rightarrow \Theta(\frac{1}{2} n \log n)$
 $= \Theta(n \log n)$

Answer: $\Theta(n \log n)$

```
int function2(int n) { NO way the function can stop early, so the best is the worst
    int count = 0; runs in constant time,  $C_0$ 
    for (int i = 1; i * i * i * i <= n; i++) { runs  $\sqrt[4]{n}$  times
        count++; runs in constant time  $C_1$ 
    }
    return count;
}
```

complexity = $C_0 + \sqrt[4]{n} (C_1)$
 $= \Theta(\sqrt[4]{n})$

Answer: $\Theta(\sqrt[4]{n})$

```
int function3(int n) {
    int count = 0; runs in constant time  $C_0$ 
    for (int i = 1; i <= n; i++) { runs  $n$  times
        for (int j = 1; j <= n; j++) { runs  $n$  times
            for (int k = 1; k <= n; k++) { runs  $n$  times
                for (int z = 1; z <= n; z++) { runs  $n$  times
                    count++; runs in constant time  $C_1$ 
                }
            }
        }
    }
    return count;
}
```

complexity = $C_0 + n \times n \times n \times n \times C_1$
 $= \Theta(n^4)$

Answer: $\Theta(n^4)$

```
int function4(int n) {
    int count = 0; runs in constant time  $C_0$ 
    for (int i = 1; i <= 2^n; i++) { runs  $2^n$  times } because of break
        for (int j = 1; j <= n * n; j++) { runs  $n^2$  times } loop only
            count++; runs in constant time  $C_1$  } runs once
        }
        break;
    }
    return count;
}
```

Complexity = $C_0 + 1 \times n^2 \times C_1$
 $= \Theta(n^2)$

Answer: $\Theta(n^2)$

```

int function5(int n) {
  int count = 0; runs in constant time  $c_0$ 
  for (int i = 1; i <= n && i < 1; i++) { since loop does not execute  $O(1)$ 
    count++;
  }
  for (int j = 1; j <= n; j++) { runs  $n$  times
    count++; runs in constant time  $c_1$ 
  }
  return count; runs in constant time  $c_2$ 
}

```

Answer: $\Theta(n)$

Complexity = $c_0 + O(1) + O(n) \times c_1 + c_2$
 $= \Theta(n)$

For functions 1-5, there are no early exits within the functions. Therefore, the most appropriate notation to use for each is Θ since the best case is the worst is the worst case